



A review of the resource efficiency and mechanical performance of commercial wood-based building materials

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ABSTRACT

With growing advocacy for structural wood-based materials in urban construction, the pressure on a sustainable supply of round wood will significantly increase over the next decades. To prevent severe material shortages, the available raw material needs to be processed as efficiently as possible. To do so, in-depth knowledge about the process efficiencies of first round wood breakdown and final material yields of established engineered wood products is needed. Therefore, this review gives insight into the process efficiency of the three dominating primary disintegration processes sawmilling, veneer peeling and log stranding as well as the material yield of subsequently produced structural engineered wood products. Furthermore, the resulting mechanical performance of the various materials is presented and a so-called “mechanical efficiency” is introduced, which combines material yield with mechanical performance. Based on the reviewed literature it can be concluded that a resource efficient combination of processes will be required in order to satisfy the increasing demand for wood-based building materials. Especially the use of veneer- and strand-based products should be intensified as they offer improved mechanical efficiency.

1. Introduction

Wood is one of the oldest structural building materials offering great mechanical performance in relation to its weight [1]. While it was predominantly replaced as a building material by steel and concrete throughout the 20th century [2] it is now poised for becoming the sustainable solution of the building industry once again [3,4]. Especially modern engineered wood products (EWP) like glue laminated timber (GLT) or cross laminated timber (CLT) are currently often used [5] due to their good technical performance [6,7] to contribute to a transition towards climate friendly cities. This is particularly important for the urban building sector, which seeks to provide a sustainable living space for the projected additional 2.5 billion urban residents by 2050 [8]. According to a survey by the OECD [9] the global construction sector will dominate the total material use by 2060. Regarding wood for construction, the additional demand could noticeably exceed the current

worldwide round wood supply (1.98 billion m³ industrial and 1.92 billion m³ fuel wood) of about 3.9 billion m³ [10–12]. Therefore, the provision of additional living space alone will require an enormous increase in productivity of the wood-based sector [13] and a re-direction of raw material from short term (e.g. energy) to long term use cycle (e.g. structural building products, reuse, recycle) [14,15]. Besides a sustainable increase in productivity and re-direction from wood for fuel to a material first approach also the efficiency of the main and subsequent disintegration processes of the wood industry needs to be addressed [16]. Based on assumptions made by Churkina et al. [12,17], 50% of the raw material is “wasted” during production and can only be used for non-structural applications (e.g. particle- and fibreboard-based furniture, insulation material, pulp and paper) or as energy source, with energy having the least positive effect on the environment [18]. This simplification provides only a rough estimate and in part ignores the wide range of processes and factors influencing the resource efficiency

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of established engineered wood products (EWP).

Additionally, changing climatic conditions will significantly influence the tree species distributions of the future [19]. Especially European forests are already severely affected by the changing climate. With the climate scenarios Representative Concentration Pathways 4.5, 6.0 and 8.5, even more drastic changes are to be expected [20]. According to findings by Hanewinkel et al. [21] the already started shift towards drought-adapted hardwood species will gain even more traction in the coming years. This will pose additional pressure on the wood supply chain of industrial nations, which is pre-dominantly optimized for softwoods like spruce, pine or fir rather than hardwoods like beech or oak [22,23]. In line with this European trend, the impact of calamities caused by weather extremes, pests, forests fires as well as the increasing share of plantation wood will also have a significant impact on the available assortments. Some of these effects can already be seen in Fig. 1, with coniferous wood species accounting for two thirds of the world's supply of saw and veneer logs in 2020 [24]. Compared to 1961 this share has already shifted significantly towards non-coniferous wood species [24], indicating an increasingly more complex raw material mix for the wood industry.

In order to deal with this paradigm shift in the building industry and forest sector, knowledge about the process efficiency of current EWPs as well as knowledge about the possible influences of hardwoods and softwoods on the material yields and properties is required.

The present paper selectively reviews the process efficiency of the three dominating first disintegration processes sawmilling, veneer peeling and log stranding as well as the material yield of subsequently produced structural EWPs. The investigated EWPs are Glued Laminated Timber (GLT), Cross Laminated Timber (CLT), Laminated Veneer Lumber (LVL), Plywood (PLY), Laminated Strand Lumber (LSL) and Oriented Strand Board (OSB). The review aims to quantify the material yield based on round wood input of established EWPs. Additionally, an attempt shall be made to analyse the differences between hardwoods (including tropical wood species and bamboo) and softwoods as raw materials. As both, the feed material and the process affect the properties of the gained products, the resulting mechanical performance of the various materials will be discussed as well and an attempt is made to combine the process efficiency with the mechanical performance.

2. Relevant definitions for this review

2.1. First disintegration and intermediaries

As depicted in Fig. 2 the first disintegration of round wood can be

categorized into sawmilling, veneer peeling and stranding [25]. These respective processes produce lumber, veneers and strands as intermediaries. Most of the commercially established structural EWPs are manufactured using one of these intermediaries [26]. Additionally, crushing or splitting [27,28] of logs into “macrofibres” as intermediaries is currently a niche process and while products like TimTek [29] are commercially available, current market shares are negligible. Therefore, this first disintegration as well as related products were not considered any further.

2.2. Engineered wood products (EWPs)

In the context of this review, EWPs are defined based on Guss [30] as products made from round wood, which has been reduced into smaller pieces (lumber, veneers or strands) and then are re-adhered together, usually using some kind of adhesive, to produce a structural material with definable mechanical properties.

Generally, GLT and CLT are mainly produced using lumber (sawn timber [31]) from the sawmilling process, while LVL and PLY are using veneers (a sheet of wood thinner than 7 mm [32] but typically 2–4 mm thick [33]) from the peeling process and LSL and OSB are manufactured using strands (a wood particle with a length of 15–300 mm, a width of 5–25 mm and a thickness of 0.2–1 mm [25,34]) from stranded round wood (see also Fig. 2). Furthermore, GLT, LVL and LSL are produced as beams, while CLT, PLY and OSB are usually manufactured as panels. This also means a similar target application within the two groups. Therefore, the EWPs could also be grouped based on the targeted use. However, as the first disintegration process clearly dictates the overall resource efficiency, a comparison based on the used intermediaries is more useful in this review. A detailed description of the production processes and the resulting products can be found in various publications [33–38]. However, some of the main characteristics of the investigated EWPs will be summarized in the following section.

GLT can be traced back to 19th century England as well as to a patent of Otto Hetzer from the early 20th century and is therefore one of the oldest engineered wood products [26]. GLT consists of at least two essentially parallel layers of, usually strength graded, lumber [39]. The individual layers are made up by smaller pieces of lumber which are jointed together using finger joints. This theoretically enables an indefinite range in producible sizes, only limited by handling and transport. Today, GLT is regulated for example in EN 14080 [39] and widely used in the residential as well as commercial construction sector.

CLT was developed by the central European wood industry in the

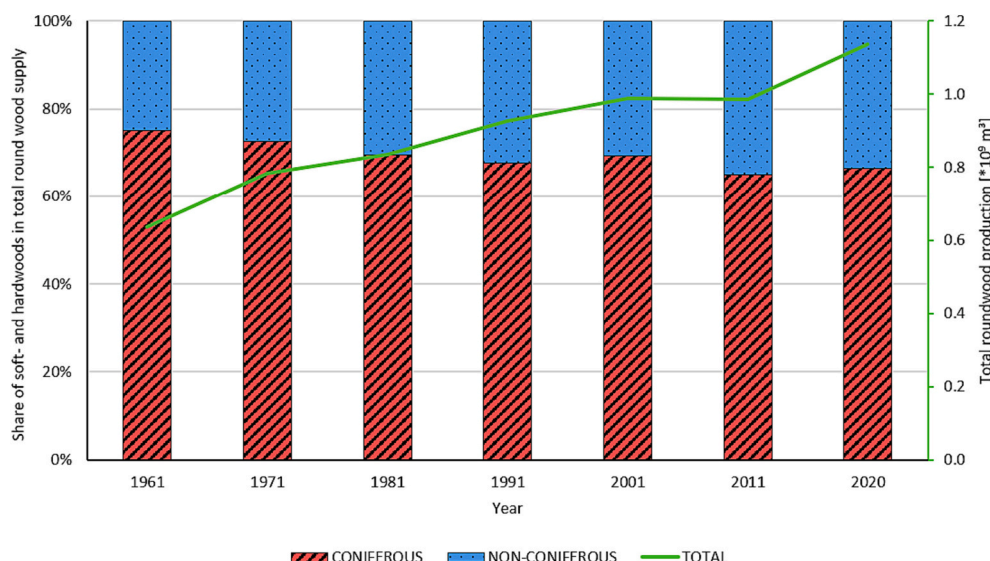


Fig. 1. World supply of saw logs and veneer logs (1961–2020) in billion m³ and share of coniferous and non-coniferous species [24].

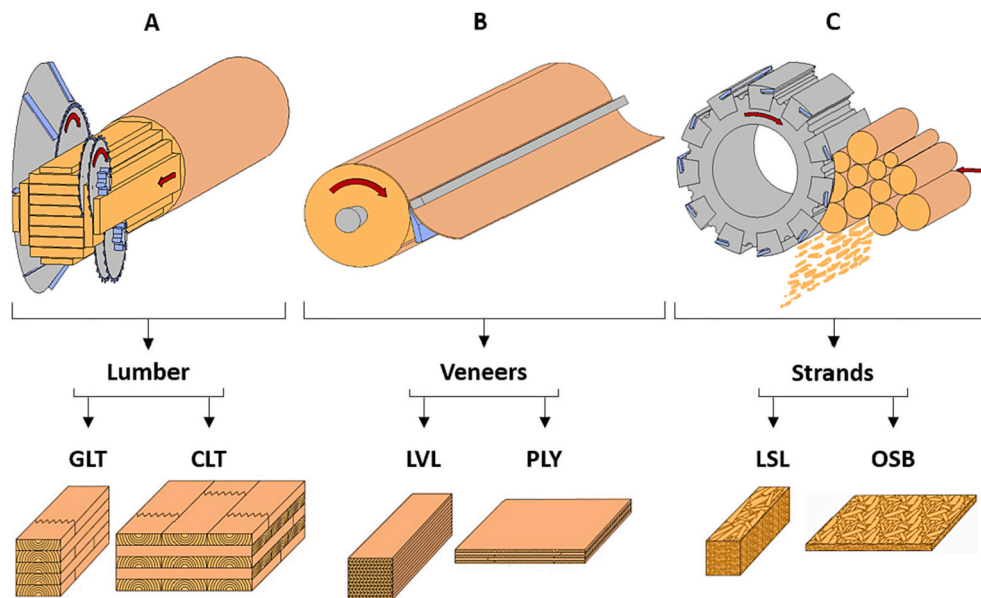


Fig. 2. Schematic depiction of main disintegration processes for round wood and subsequently produced engineered wood products relevant for structural applications. A: Sawmilling to produce lumber/timber used in GLT = Glued Laminated Timber and CLT = Cross Laminated Timber. B: Veneer peeling to produce veneers used in LVL = Laminated Veneer Lumber and PLY = Plywood. C: Stranding to produce strands used in LSL = Laminated Strand Lumber and OSB = Oriented Strand Board.

mid-1990s, which needed a use for its side boards from the sawmilling process [40]. The symmetrical lay-up and orthogonal lamination of at least three layers of lumber proposed by Schickhofer [40] laid the ground work for a, at the time, low-cost, high-value product with high throughput [26]. The cross-wise alignment leads to a high dimensional stability similar to plywood. The high degree of prefabrication and standardized dimensioning have further favoured market growth in recent years. Today, CLT as a product is regulated for example in EN 163451 [41] and production volumes have grown exponentially in the last years, reaching over one million m³ in 2021 for the first time [42].

LVL for structural purposes is mass produced since the 1970s, but was already used for airplane construction at the beginning of the 20th century [33]. It is made of peeled veneers. Which are bonded together with parallel fibre direction. In comparison to GLT, the utilization of veneers leads to a better distribution of defects within the cross section of LVL which improves the homogenization of material properties and usually allows for higher design values [43]. Today, LVL is regulated for example in EN 14374 [44] and is one of the most widely used structural EWP in the United States [36].

PLY can be seen as the oldest engineered wood product, manufactured as early as 3500 years ago through crosswise joining of sheets of wood. [26]. Back then these sheets were made out of sawn wood and were held together with dowels. With the introduction of rotary veneer lathes in the 19th century and thermosetting adhesives in the 1960s, the high-volume production of structural plywood, now based on peeled veneers, became possible [33]. Today, PLY is regulated for example in EN 636 [45] and is the most important veneer-based EWP globally [46].

OSB and subsequently LSL were developed in the mid of the 20th century. The main goal was to find a substitute for relatively expensive plywood by utilizing strands produced from smaller diameter round wood, which was cheaper than the high diameter round wood necessary for veneer production [33]. OSB is usually comprised of three layers, with the strands in the surface layers being somewhat orientated parallel to the major axis and the strands in the core layer perpendicular or (mostly) random to it [26]. In contrast to OSB, the strands in LSL are all oriented parallel to the major axis and also show a higher degree of alignment [36]. Today, OSB is regulated for example in EN 300 [47] and LSL for example in EAD 130308-00-0304 [48].

It should also be mentioned that historically dominating EWPs like

Nail Laminated Timber (NLT) or Dowel Laminated Timber (DLT) [49] will not be considered in the following comparison. While their importance may significantly increase in the future due to their good recyclability, they are currently only available to a limited extend [50,51] and are not generally regulated through standards. However, as both use dimensional lumber as base intermediary it can be assumed that they are probably in the same ranges as GLT or CLT with regards to resource efficiency.

2.3. Resource efficiency

Resource (process) efficiency, also referred to as material or product yield, is the ratio between the output of intermediate or final products and the input of logs or intermediaries and is commonly calculated based on the following equation:

$$\text{Efficiency or yield} = \frac{\text{Intermediate material or product output}}{\text{Logor intermediate material input}} \times 100 [\%] \quad (1)$$

To calculate the yield of intermediate products, processing losses [36] during drying and planing in the case of sawmilling, drying and clipping in the case of veneer peeling and drying and screening for non-useable fines in the case of stranding are subtracted. The subsequent production of structural EWPs causes additional losses [36]. In the case of GLT and CLT finger-jointing, planing/sanding and end cutting of lamellas are the main drivers influencing the final yield. Regarding LVL and PLY the losses due to defect removal, formatting of the finished panels and sanding of the final panel are leading to additional losses. Considering LSL and OSB final yield is mainly determined by formatting and sanding. It is important to note here that these so-called losses can be distinguished in waste which is directly disposed of and by-products like sawdust and wood chips which can be used for products like particle and fibre boards. However, these products usually offer lower added value and are almost exclusively used for non-load-bearing applications. Therefore, the yields discussed below do not differentiate what is done with these losses, but rather compare efficiencies in the production of long-lasting, load-bearing components.

In regards to true round wood input the measuring principle (volume

or weight) as well as the consideration of the bark content dictates how one unit of round wood input is defined and therefore significantly influences the reported efficiency [52,53]. According to the Food and Agriculture Organization of the United Nations (FAO) [54] different national log scales show a deviation from true log volume as high as 32.3% with an average of 8.4%. Additionally, when converting 1000 Board feet (traditionally used in North America) to SI volumes, calculated cubic meters can range from 3.87 m³ up to 17.23 m³ depending on the log scale and small end diameter class [54]. These inconsistencies need to be kept in mind when comparing data on process efficiency and yield of EWP.

3. The literature on resource efficiency

Literature dealing with the process efficiency of round wood breakdown and yield of EWPs are usually focusing on either Life Cycle Assessment (LCA) [55–58], mathematical modelling [59–62], process technologies [63–65] or national [66,67] and international [54,68] reporting of material flows. A comprehensive list of relevant publications is provided in the appendix (see Table A). For further information on the discussed log and lumber characteristics the reader is referred to ISO 24294 [69] for definitions. The efficiency reported in the literature is based on different units of measurement which is also indicated in Table A. The most common terms to describe the process efficiency of sawmilling are Lumber Recovery Factor (LRF) and Lumber Recovery Ratio (LRR). The yield of veneer peeling is mostly reported with the

Veneer Recovery Factor (VRF) or Veneer Recovery Ratio (VRR). A detailed description of the terms was done by Fahey [70]. Subsequent production of lumber and veneer-based EWP is often reported based on m³ intermediary input and m³ EWP output. Additionally, efficiency is also reported based on the dry weight of log or intermediary input and EWP output. In some cases, no specific method for calculating efficiency is described, and yield is essentially reported as a percentage of raw material input. This literature is indicated with a “% from log” basis in Table A and is comparable with LRR and VRR. In general, the described factors are calculated based on wood input without bark (log under bark). However, logs are mostly delivered to mills with bark and are usually debarked as one of the first processing steps. Based on the cases where bark content is reported in Table A, on average 10% of the biomass delivered to the mill is bark. It is important to note here that bark is usually subtracted by a fixed proportion or a fixed diameter reduction when measuring log volume according to wood species and diameter. This results in a potential inaccuracy in the actual amount of bark and the actual amount of wood input.

3.1. First disintegration and intermediaries

The reported values for sawmilling and veneer peeling as well as the total yield for OSB are summarized in Fig. 3.

OSB was chosen as representative for log stranding as there is little literature available [71,72] on the efficiency of mere log stranding. Since only the fines are regarded as losses and the strands are not a

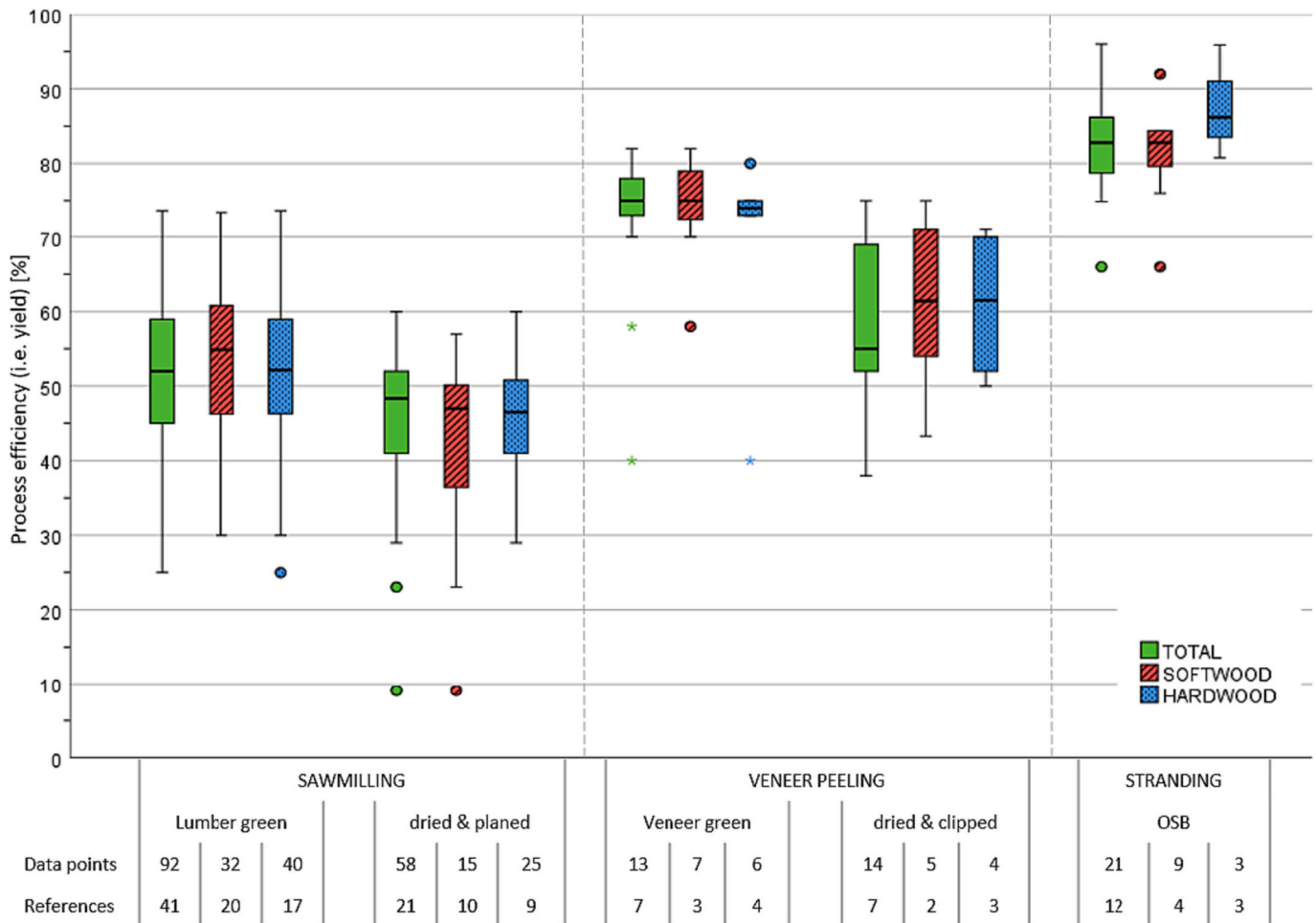


Fig. 3. Process efficiency [%] of the first round of wood breakdown and subsequent processing losses until the final intermediary product and total OSB yield as reference for the stranding process. The base input material is round wood under bark (excluding bark), with the input unit being either cubic volume or weight. (Sources: Table A).

classic sellable intermediary, OSB can be considered a good representation of the stranding process itself. Especially as the further processing losses during OSB production can be considered comparably low based on the process reported in the literature [26,36]. Based on 41 references (see Table A) the average total green lumber yield related to logs under bark ranges from 25% up to 74% with an average of 52%. In general, sawmilling efficiency increased due to improved process technologies over the past decade's [73–76]. Nevertheless, various factors influence sawmilling efficiency and are summarized in Table 1.

Market characteristics (e.g. price of lumber vs. price of by-products) influence sawmilling efficiency by conscious decisions against or in favour of an improved lumber yield [63,77]. In contrast, mill, log, process and lumber characteristics are pre-dominantly discussed against the background of improving lumber yield. In regards to log characteristics, diameter, length, taper and curvature are the most important factors influencing process efficiency. Based on findings in the literature [52,60,63,70,80] lumber yield increases with increasing log diameter, while yield generally decreases with increasing length, taper and curvature. To compensate for differing log shapes and sizes, process characteristics like sawing patterns (opening faces) [52,63,83,87], as well as rotation and positioning [88,89] of the log are usually a direct answer to raw material characteristics to maximize yield on a log-to-log basis. Additionally, general equipment properties like machine set up [60,65] and kerf width [52,60,63] along with maintenance schedule [63] also influence the process efficiency. However, the ongoing improvements in 3D X-ray scanner technology have enabled the industry to obtain information about the internal structure of the raw material and have led to a shift towards value-focused processing rather than yield based [97].

Considering the next processing steps of green lumber, the literature reports an average total dried & planed lumber yield from 9% up to 60% with an average of 46% based on 21 references (see Table A). The lowest reported dried & planned lumber yield of 9% [57] is based on an assumed distribution of process efficiencies and very likely reflects only a theoretical worst-case scenario. Similar to round wood volume, the methods of determining lumber volume influence reported process efficiency. Most commonly, actual dimensions of lumber are used to determine the cubic volume (length x width x height). One exception can be found in North America, where nominal dimensions are used to calculate board feet (bf) of lumber [52,54,73]. One board foot is defined as a 1-in. (2,54 cm) thick and 12 by 12 in. (30,48 cm) long board, which would lead to 1 m³ being 424 bf [52]. However, as these nominal sizes sometimes ignore dimensional changes due to drying and planing of the lumber actual volume could be as low as 0.66 m³ [54]. Besides these variabilities, drying (6% average drying loss [54,57,70,77,91,98–102]), planing (13% average planing loss [54,57,67,70,77,91,98–104]) and sorting [77,91–96] significantly decreases final intermediary yield of the sawmilling process (also see Table 1).

Moving on to veneer peeling, the total green veneer yield ranges from 40% up to 82% with an average of 72%, showing low variability within the 6 references (see Fig. 3). Similar to sawmilling, technological advancements increased veneer recovery over the last decade's significantly [36]. The chipless peeling of veneers leads to a generally higher material yield compared to lumber from sawmilling [37] and similarly

veneer yield increases with increasing small end diameter [105]. However, this requires uniform raw material of preferably high diameter and high quality [36,105]. In addition, the machine configuration also plays an important role. Currently, spindleless lathe technology is the state of the art [106,107], enabling the log to be peeled down to a residual diameter of about 50 mm [36].

After peeling, green veneers are usually clipped and dried to target moisture content. Total dried & clipped veneer yield ranges from 38% up to 77% with an average of 57% showing higher variability (Fig. 3) compared to green veneers, based on 6 publications (see Table A). Comparable to the sawing process, there are also different ways in which the veneer volume is determined. The cubic volume is either measured based on actual dimensions (length x width x thickness) or based on the surface area on an equivalent thickness basis [54]. For example, a veneer with 1.25 m² and a 1 mm basis would equal 0.00125 m³, with the basis being the actual target thickness including a small plus or minus tolerance [52]. As both methods are based on actual dimensions the variability in reported veneer yield should be negligible. More importantly, the loss due to clipping is reported [52,70,105,108] to be the most severe factor in decreasing veneer yield. According to findings by Woodfin [105] as well as Fahey and Woodfin [70] roughly 20% of material is turned into by-products, whereby the clipped volume (mainly cracked and defective areas) is significantly influenced by the applied clipping strategy [109]. Final veneer drying further reduces the overall yield, with the target moisture content ranging from as low as 2% up to 15% [36] usually dictated by the target size [52] and possible bonding conditions for subsequent LVL and PLY production [34].

Taking a look at log stranding, OSB yield ranges from 66% up to 96% with an average yield of 83% based on 12 references. In contrast to sawmilling and veneer peeling, the log stranding process is mainly monitored based on weight. However, final products are mainly reported in cubic volume or surface area on a thickness basis similar to veneers [54]. Since moisture content and compaction during panel pressing are considered based on conversion factors, there may be certain disparities in reported process efficiencies. Nevertheless, the log stranding process does not require high-quality, mature logs and is rather geared towards small-diameter round wood of lower quality from thinnings or plantations [26]. Therefore, the process allows for a bigger range in raw material quality and enables more flexibility with regard to diameter and shape. The factors influencing strand yield partly differ from the previous processes. Increasing log moisture content [71] and ambient temperature [72] significantly decreases the produced fines in stranding and therefore increases strand yield. These fines are normally separated from OSB production as they reduce the strength of the final board and increase the amount of adhesive required. In regards to machine set-up, the knife angle is reported [72] to be an important factor influencing the share of fines. In addition, the positioning of the logs against the rotating knife by means of clamps or pushers partly dictates the strand geometry and therefore also influences the strand yield [36]. The final target moisture content of the strands is in the range of 2–5% [26,36] usually dried from above fibre saturation [110]. As mentioned previously, this could further affect the final material yield depending on the reported input and output unit. This is especially critical when

Table 1
Factors influencing sawmilling efficiency.

Category	Factors	References
Market characteristics	Product mix, prices, demand and price for by-products, round wood supply	[52,63,75,77]
Mill characteristics	Production capacity, equipment type, age of the machinery, decisions making	[63,65,73–75,78]
Log characteristics	Species, origin (native or plantation forest), diameter, length, taper, curvature, knottiness, position of crosscut	[52,60,63,70,73–75,79–86]
Process characteristics	Cutting pattern, accuracy, saw kerf, rotation and positioning, curve sawing capabilities, condition and maintenance of equipment	[60,61,63,75,77,83,87–89]
Lumber characteristics	Species, target size (nominal vs actual), moisture content (drying), visual characteristics (twist, bow, knots, ring width, checks, wane), applied sorting criteria	[52,59,63,77,82,85,90–96]

comparing the different processes, as volumetric shrinkage is usually considered a loss in volume-based lumber and veneer production. However, as OSB production is commonly monitored using dry weight, this could lead to inaccuracies when comparing the processes.

Based on Fig. 3, the influence of wood type (hardwood vs. softwood) seems to be negligible for the first round of wood breakdown. This underlines the greater influence of the principal technology on the level of achievable resource efficiency as compared to the processed wood species. Considering saw milling and veneer peeling, the optimization of the mills for either softwood or hardwood processing is reported [63,111] to be a key factor influencing the process efficiency. This emphasizes the importance of a planable and steady supply of comparable round wood assortments for these industries. Currently, the smaller share of hardwoods processed by the wood industry (see Fig. 1) is either subjected to a stricter degree of pre-sorting compared to softwoods, with the non-sawable and -peelable assortments being either used for energy supply or pulp and paper, or is directly felled for these low added value sectors [24]. Therefore, the attainable round wood quality of hardwoods will be a key issue in the future [21,112] and could significantly reduce the process efficiency of sawmilling [86] as well as veneer peeling [111].

Furthermore, a closer look at other process-related factors like productivity, energy demand, necessary pre-treatment and resulting intermediary quality is needed to get a full view of the effect of the changing raw material situation. A quick look into the influence of intermediary quality on the final material yield of lumber and veneer-based EWPs is taken at the end of the next chapter.

3.2. Engineered wood products (EWPs)

So far, mainly the first disintegration of round wood into intermediaries has been discussed. The following section focuses on engineered wood products relevant to building applications. In regards to GLT and CLT production, the literature consistently starts with lumber that is either dried or dried and planed. Similar is the case with LVL, where literature starts with dried and clipped veneer sheets, although there are also publications that start with logs. Therefore, the reported minimum, maximum and average intermediary yield (dried & planed lumber as well as dried & clipped veneer) was multiplied with the average reported further processing efficiency to enable a comparison of relevant EWPs in Fig. 4.

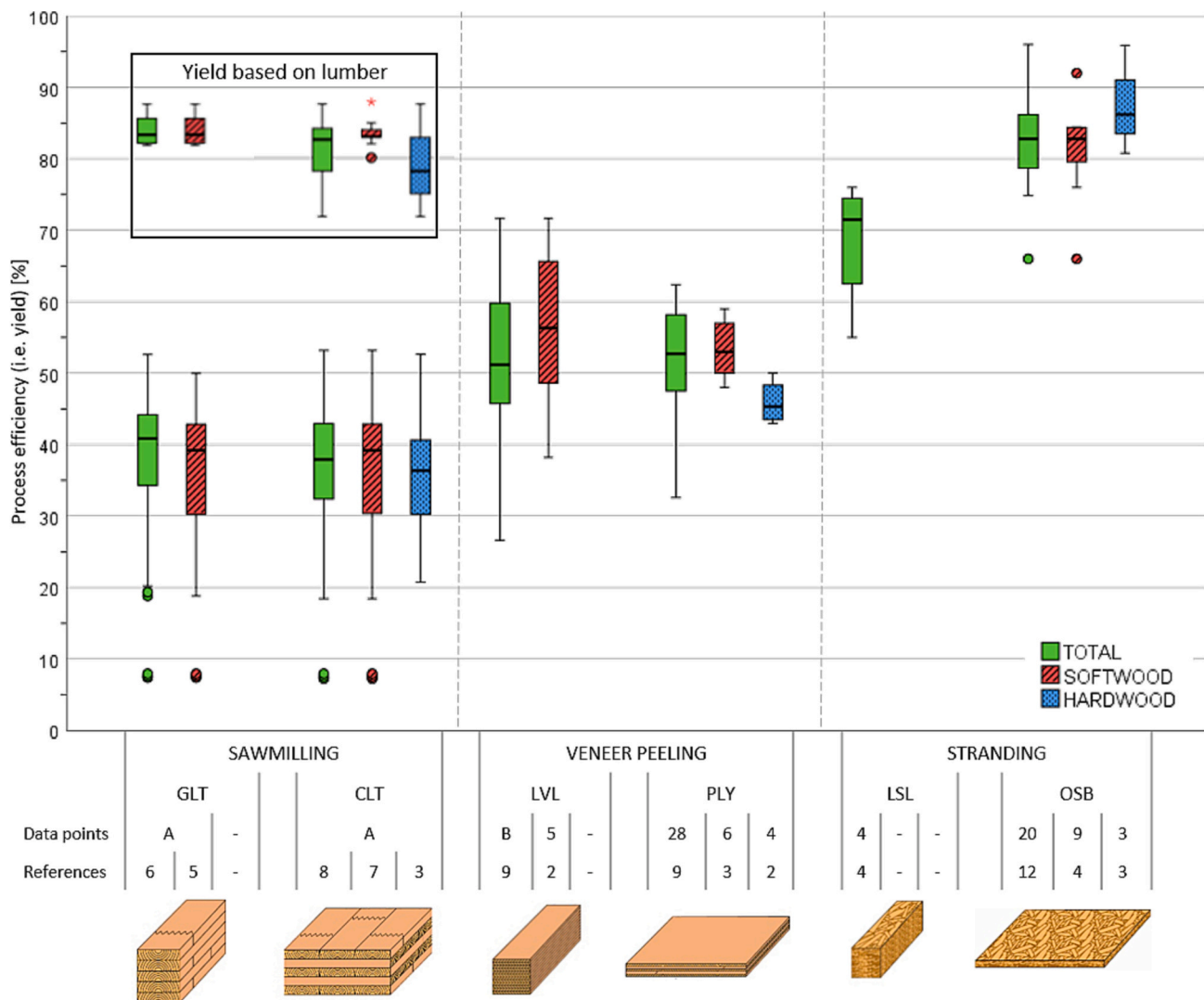


Fig. 4. Material yield [%] of selected engineered wood products based on round wood under bark (excluding bark). A: Yield for GLT, CLT is based on the minimum, maximum and average process efficiency of lumber dried & planed (see Fig. 3) multiplied by the average further processing loss reported in the literature. B: Yield for LVL is based on literature and the minimum, maximum and average process efficiency of veneer dried & planed (see Fig. 3) multiplied by the average further processing loss reported in the literature. Further processing efficiencies for the production of GLT and CLT are also reported based on either dried or dried and planed lumber input (see box). (Sources: Table A).

Taking a look at the lumber-based products GLT and CLT, comparable efficiencies can be observed, with total GLT yield ranging from 7% up to 53% and an average of 39% and total CLT achieving an average yield of 37% ranging from 7%–53% based on 6 and 8 references respectively. Overall, the further processing losses of GLT and CLT are rather similar within different literature (see the box in Fig. 4). For total GLT [58,113–115], the average losses during production amount to about 16% in a range of 12–18% based on lumber input. In regards to CLT [57,62,104,115–119], roughly 16% of input material is lost, ranging from 12% up to 20%. From this it can be argued that the main reason for the high variability in yield can be found in the base efficiency of the input lumber. In both cases the main steps causing additional losses are lumber trimming and finger jointing, planing of the lamellas and finishing of the panels [26,120]. However, it is important to note that especially planer shavings, which are a big loss factor [57,76], are probably counted double in some cases for GLT as well as CLT as the process may also be started with rough-sawn lumber [116]. For CLT, this is countered by the fact that the losses caused by the joinery (e.g. cut-outs for windows and doors) are not yet included. Depending on the end-use of the panel this could further bring down the final yield significantly. While floor panels do not require much cut-outs, outer and inner walls require openings for windows or doors, with the total area of these openings mainly driven by design aspects as well as building physics (lighting and ventilation) [121,122].

Moving on to the veneer-based products, LVL and PLY achieve higher average yields compared to GLT and CLT. The total LVL yield ranges from 27% to 72% with an average of 52% based on nine references, with an average further processing loss of 9% [123–125]. For total PLY, an average yield of 52% in the range of 33%–62% can be reached according to 9 references. For the final volume determination, the same arguments apply as for veneer and OSB. Additional losses are caused by waste veneer, lay-up scrap, panel trimming and sanding as well as material testing [52,70,123,124,126], with improved material utilization of roughly 15% compared to the lumber-based EWP. The higher maximum yield of LVL compared to PLY could be explained by fewer cross cuts and fewer surfaces for sanding as usually thicker billets are produced [127].

Concluding the EWPs, the strand-based LSL and OSB achieve the highest material yield, with total LSL ranging from 55% up to 76% having an average of 69% and total OSB using an average of 83% of one unit of log input in a range from 66% to 96% according to 4 and 12 publications respectively. In regards to LSL, the generally lower yield compared to OSB is due to the unidirectional structure over the entire thickness of the material, demanding more uniform strands, with high aspect ratios [128]. Therefore, part of the finer fractions which are usually put in the middle layer of OSB cannot be used for the homogeneous build-up of LSL beams. Besides these differences, the main drivers for additional material losses are formatting and sanding of the finished panels and beams [26,126,129].

Depending on the EWP (also see Fig. 4) no or too little literature was available to make a general statement regarding the influence of wood species on the final material yield. Taking a look at the lumber-based products GLT and CLT, there was no literature on the processing losses of GLT made from hardwoods and the literature found for CLT is primarily based on the same numerical simulation of low-grade yellow-poplar round wood [59,130] reporting a lower yield compared to softwoods. In addition, the losses through pre-sorting and strength grading of the input lumber are reported to be significantly higher for hardwoods [77,91] than for softwoods [92,93]. In regards to LVL and PLY, a similar picture unfolds with no literature available on hardwood LVL and little literature on PLY [54,131], showing values significantly lower compared to softwoods. In addition, total LVL and PLY includes literature [52,70,74,124,126] reporting a mixture of both soft- as well as hardwoods, exhibiting a higher variability in yield compared to mere softwoods. This indicates that the higher inhomogeneity of hardwood tree species also influences the range of achievable material yields of

LVL and PLY. In contrast to the lumber and veneer-based EWPs, Fig. 4 shows a different picture for OSB, with hardwood OSB having a higher material yield compared to OSB made from softwoods. In that regard, results from Feng and Knudson [71] show a significantly higher share of fines from green pine (15.4%) compared to green aspen (4.1%). This indicates, that the production of hardwood OSB could result in a generally higher material yield. However, a more detailed look at other relevant hardwood species is necessary to confirm this statement based on a single study. As aspen can be considered a low-density hardwood species [132] with technological properties similar to softwoods [133], this statement cannot be directly transferred to other hardwoods (e.g. oak or beech).

4. Combining resource efficiency with mechanical performance

So far EWPs have been discussed against the background of material yield. However, the architectural freedom [134] as well as the desired number of floors [135] in buildings, such as the ongoing trend towards tall buildings demands a certain mechanical capacity of the possible construction materials [136]. In the case of EWPs, the mechanical performance is typically based on a combination of various raw materials, intermediate materials and process-related factors, which influence the corresponding resource efficiency. Additionally, the lower process efficiencies of lumber, as well as veneer-based hardwood EWPs, could be balanced by the higher attainable strength classes compared to softwoods [44,137]. Therefore, the mechanical performance of the presented EWPs in relation to the respective material yield is of special interest and is discussed in the next section.

4.1. Selected mechanical properties

Based on Eurocode 5 [6] two structural checks are mainly required during the design phase of statically loaded building components. While the ultimate limit state (ULS) ensures the structural integrity of a building, the serviceability limit state (SLS) ensures that predefined terms of use like the deformations or vibrations of a structure remain within defined limits. With that in mind, two material properties were chosen as being representative for each of the design states. The bending strength indicates the maximum stress a material can carry under flexural loading and is therefore a suitable benchmark when it comes to the ULS design of a building member loaded in bending. The modulus of elasticity, on the other hand, represents the capacity of a material to resist deflection when subjected to a load and is therefore a key material property when it comes to the SLS design of a building member. Additionally, the compression strength and compression stiffness are of equal interest, especially in the case of walls and columns of a building. However, both properties are not as frequently available as bending strength and stiffness. Therefore, they are not considered in the subsequent comparisons but they could be of greater importance in a real design process. Furthermore, each load-bearing element within a building fulfils a specific structural function, which influences its geometry and degree of anisotropy [138]. Based on structural function, the discussed EWPs can be grouped one-dimensional elements like beams (loaded normal to the axis), columns (loaded parallel to the axis) and two-dimensional elements like walls (loaded in-plane) and plates (loaded normal to the plane). Additionally, material properties are usually given for the main (0°) as well as secondary (90°) load direction. Based on these considerations, the characteristic bending strength ($\sigma_{B0, \text{char.}}$) and the mean modulus of elasticity ($E_{B0, \text{mean}}$) in the main load direction as well as the average mean density (ρ_{mean}) found in standards and technical approvals are summarized for further comparison in Table 2.

It needs to be mentioned that, the actual strength may be significantly higher on a piece-by-piece basis. This is especially true for the lumber-based EWPs, which commonly exhibit a high variability of actual strength values due to the natural inhomogeneity of wood

[132,179]. In addition, the safety factors for determining the design values differ depending on the product, the load case and the respective standards [6,7179]. In regards to CLT also the currently used dimensioning approaches need to be discussed. As the transverse layers are not considered when deriving the net cross-section during the design phase [6,7120], the section modulus is reduced compared to a similar cross-section with a unidirectional build-up and hence the applied stresses are increased. Depending on the layer structure of the CLT panel this leads to a reduced load-bearing capability. Therefore, the tabulated characteristic bending strengths should be seen as an indicator of the baseline mechanical performance.

4.2. Characteristic bending strength ($\sigma_{B0,char}$)

Regarding the characteristic bending strength in Table 2, a comparable level of performance for the investigated EWPs can be seen. However, the significant influence of the inhomogeneity of commonly processed timber on the permissible strength of the resulting EWPs can be seen when the possible bending strength of defect-free wood is considered [133,180,181]. This can be explained by the influence of defects [133] in combination with a size effect [182–184]. The influence of defects in larger pieces of wood is more detrimental to the permissible characteristic strength. This is especially critical for GLT and CLT which underperform significantly compared to the other products, particularly when process efficiency is added to the discussion (see Fig. 5). Both show the lowest average material yield (39% for GLT and 37% for CLT) while only allowing for similar or even lower maximum stresses as the veneer and strand-based products. However, Fig. 5 also shows that the potential maximum strength can be significantly higher than for LSL and OSB while allowing for comparable values to the veneer-based EWPs.

Moving on to LVL and PLY, both offer a good balance between mechanical performance and resource efficiency. LVL showed the highest average characteristic bending strength ($\sigma_{B0,char} = 73,5$ MPa for hardwoods) of all investigated EWPs while achieving comparably high material yields with up to 72%. PLY on the other hand does not offer quite as high mechanical performance as LVL, exhibiting 22,4 MPa for softwoods and 42,2 MPa for hardwoods. This can be explained by the directionally balanced material performance due to the transverse layers and the higher influence of process-related micro cracks [36,185] present in veneers. However, mechanical performance is still comparable to CLT while offering a potential higher material yield of around 52%.

In regards to the strand-based EWPs, LSL and OSB offer practicable characteristic bending strength (32,7 MPa for LSL and 20,4 MPa for OSB) while achieving the highest material yields of the discussed composites. Especially the approved but not available products like EUROSTRAND® OSB 8000 [178] and TimberStrand® [165] show the high mechanical potential of strand-based EWPs, with a characteristic bending strength of up to 35 MPa and 57 MPa respectively. Similar to plywood, the crosswise build-up of OSB, with the face layer strands

being oriented in panel direction and the core layer perpendicular to it, leads to trade-off in strength compared to the unidirectional LSL. However, compared again to the also panel-shaped CLT, OSB remains mechanically competitive while achieving twice the resource efficiency (83%) on average.

4.3. Specific bending strength (spec. $\sigma_{B0,char}$)

As stated in the previous chapter (4.2.), hardwoods offer the possibility to achieve higher strength classes compared to softwoods. Based on the mean characteristic bending strength in Table 2, hardwoods show an improvement in strength of about 49% for GLT, 78% for CLT, 94% for LVL and 88% for PLY. However, this increased mechanical performance usually comes at the price of heavier products due to the higher density of hardwoods being one of the main reasons for the increased strength [133]. For further comparison, this difference is considered by dividing the characteristic bending strength of the respective product by the corresponding mean density. Furthermore, the resulting specific characteristic bending strength (spec. $\sigma_{B0,char}$) is displayed next to the achievable material yield in Fig. 6.

The average specific characteristic bending strength (spec. $\sigma_{B,char}$) of GLT made from softwoods is 0,053 Nm/kg and 0,054 Nm/kg for hardwoods, indicating that both offer the same strength-weight ratio. For CLT (0,040 Nm/kg for softwoods) this comparison is not useful since there is only data on one hardwood product available [150]. Regarding LVL (0,069 Nm/kg in the case of softwoods and 0,092 Nm/kg for hardwoods) and PLY (0,045 Nm/kg for softwoods and 0,062 Nm/kg for hardwoods), the better mechanical performance cannot be fully explained by the density. The remaining difference is most likely explained by the better log quality for the veneer-based materials. Another explanation could be the improved homogeneity with fewer and better-distributed defects within the product allowing for higher characteristic strength and therefore better strength-to-weight ratio. For the strand-based products, a comparison within one product is not possible as LSL is primarily based on hardwoods (usually poplar) and OSB mainly on softwoods (usually pine), with some companies also producing OSB from poplar logs depending on availability. Between the two, however, LSL has a higher spec. $\sigma_{B,char}$ (0,047 Nm/kg) as OSB (0,032 Nm/kg) with the remaining advantage of LSL explainable by the longer, purely unidirectional oriented strands.

4.4. Modulus of elasticity ($E_{B0,mean}$)

For the stiffness of the discussed EWPs the range of mean modulus of elasticity is displayed next to the material yield in Fig. 7.

Compared to the characteristic bending strength, the modulus of elasticity shows a closer grouping of values around 11 GPa. An outlier can be found for OSB, with a minimum mean modulus of elasticity of 4,9 GPa. This significantly lower stiffness compared to the other products

Table 2

Structural function, density (ρ), characteristic bending strength ($\sigma_{B,char}$) and modulus of elasticity ($E_{B0,mean}$) of approved products and standards. Avg. = average of reported values, Min. = minimum of reported values, Max. = maximum of reported values, Ref. = Reference.

Intermediary	EWP	Function	Species	Avg. ρ_{mean} [kg/m ³]	Min. $\sigma_{B0,char}$ [MPa]	Avg. $\sigma_{B0,char}$ [MPa]	Max. $\sigma_{B0,char}$ [MPa]	Avg. $E_{B0,mean}$ [GPa]	Ref.
Lumber	GLT	beam	SW	500	11,6	26,7	43,4	12,1	[39,139–145]
			HW	733	28	39,7	75	14,1	
	CLT	plate	SW	531	6,7	21,4	49,4	8,9	[146–153]
			HW	620	–	38	–	15	
Veneers	LVL	beam	SW	545	28	37,8	54,4	12,7	[44,143,154–169]
			HW	800	54	73,5	91,7	15,3	
	PLY	plate	SW	495	14,7	22,4	37,6	8,5	[128,170–174]
			HW	680	36,8	42,2	65,9	10,2	
Strands	LSL	beam	SW	not available	–	–	–	–	[165]
			HW	698	27,5	32,7	57	11,7	
	OSB	plate	SW	631	14,8	20,4	35	5,9	[175–178]
			HW	not available	–	–	–	–	

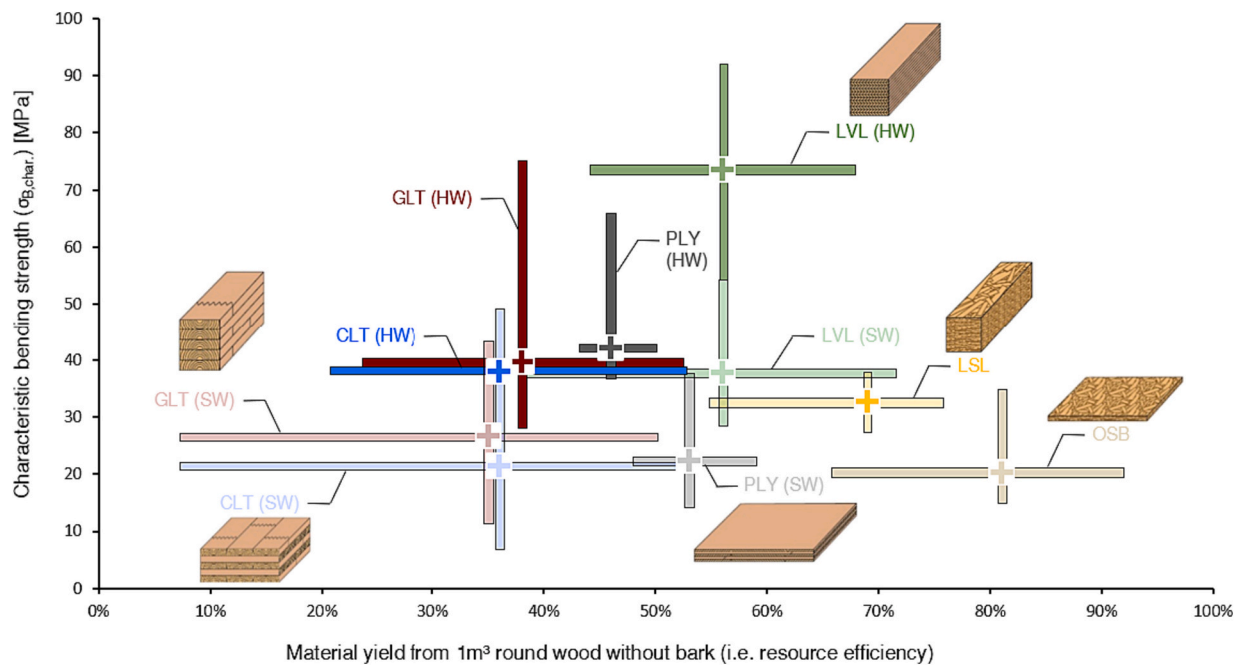


Fig. 5. Characteristic bending strength [MPa] of GLT [39,139–145], CLT [146–153], LVL [44,143,154–169], PLY [128,170–174], LSL [165] and OSB [175–178] made from softwoods (SW) or hardwoods (HW) based on values in Table 2 in relation to the achievable material yield [%] from 1m³ of round wood without bark as reported in Fig. 4 (either soft- or hardwoods).

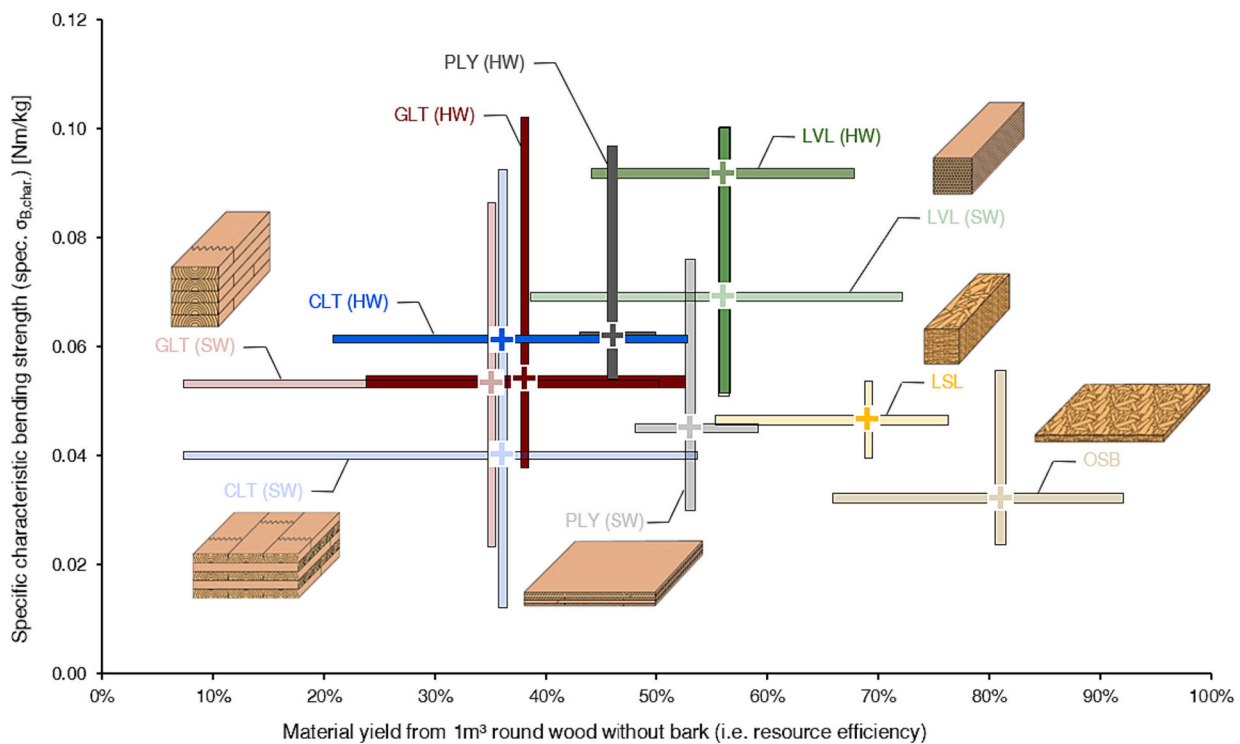


Fig. 6. Specific characteristic bending strength [Nm/kg] of GLT [39,139–145], CLT [146–153], LVL [44,143,154–169], PLY [128,170–174], LSL [165] and OSB [175–178] made from softwoods (SW) or hardwoods (HW) based on values in Table 2 in relation to the achievable material yield [%] from 1m³ of round wood without bark as reported in Fig. 4 (either soft- or hardwoods).

will cause higher instantaneous deflections and is therefore a major disadvantage of OSB when it comes to the serviceability limit state of a building member. The main reason for this can be found in the used intermediary. According to findings by Barnes [186–188] and Beck et al. [189] the stiffness of OSB is influenced by the size and orientation of the

used strands, with shorter and thicker strands as well as a more random orientation leading to a decrease in modulus of elasticity. Especially in comparison to the also strand-based, but fully unidirectionally oriented LSL, OSB only achieves around half the stiffness (see Table 2). This is a serious disadvantage of OSB for structural applications and cannot be

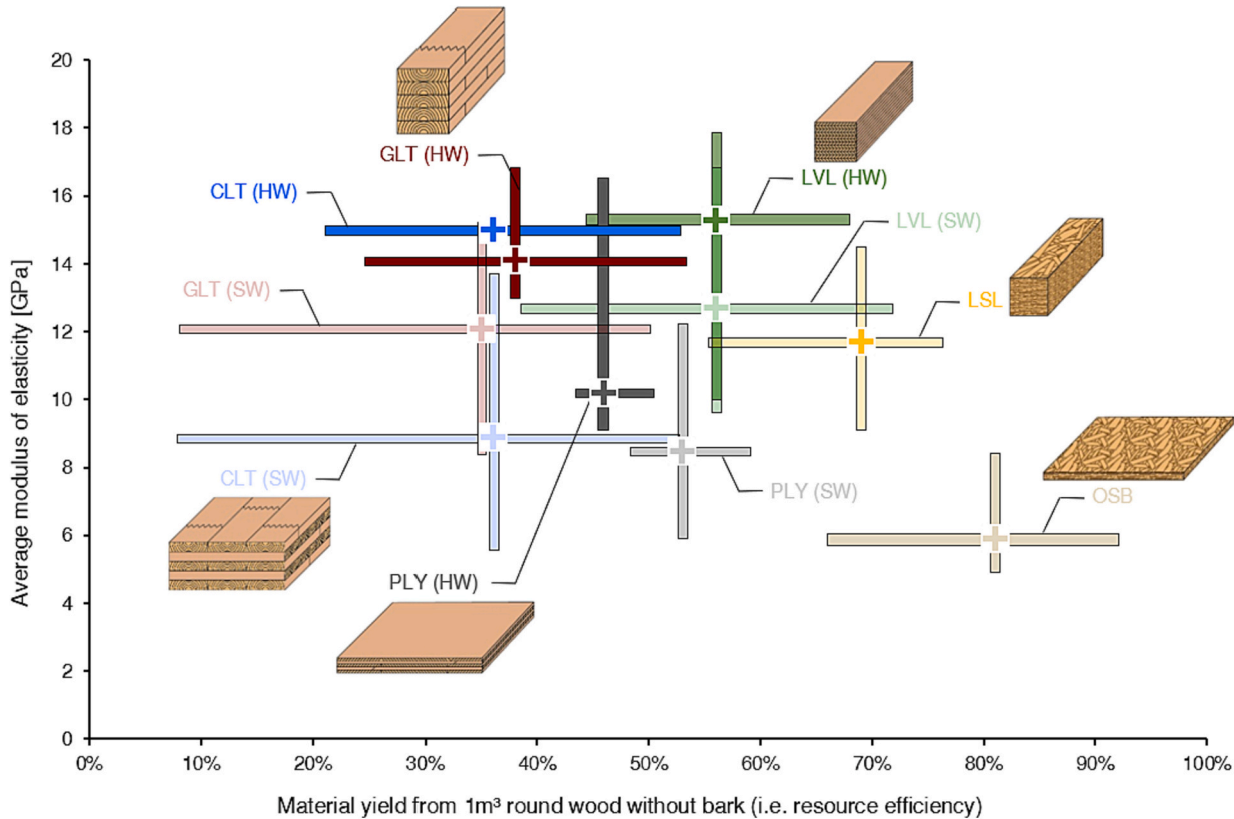


Fig. 7. Average modulus of elasticity ($E_{B0,mean}$) [GPa] of GLT [39,139–145], CLT [146–153], LVL [44,143,154–169], LSL [165] and OSB [175–178] made from softwoods (SW) or hardwoods (HW) based on values in Table 2 in relation to the achievable material yield [%] from 1m³ of round wood without bark as reported in Fig. 4 (either soft- or hardwoods).

compensated by the improved material yield. However, as part of a hybrid construction approach OSB could be used to improve the overall yield of the composite.

4.5. Long-term mechanical performance

Beside the immediate mechanical performance, the long-term usability and safety is another essential aspect for the construction sector. Especially the load duration as well as climatic conditions are particularly important in regards to wood-based materials. In general, the long-term behaviour of wood is influenced by its main components (cellulose, hemicellulose and lignin), its moisture and to some extend also its extractive content. Additionally, the size of the structural element (lumber, veneers, strands, etc.), the used adhesive and the interaction between the two also play an important role in regards to structural EWP [33]. It should be emphasized here that the long-term behaviour of wood in general as well as of EWPs is a very complex topic and that the field is so far only partially explored. We want to mention current approaches for the sake of completeness but are not treating this part as key topic of the review.

In order to account for the variety of influences on the long-term safety and usability, established design concepts like the Eurocode 5 [190] use so-called modification factors (see Table 3) to adjust the corresponding mechanical properties or loads to the respective use case.

From Table 3 it can be seen that most of the EWPs have similar long-term behaviour, except for OSB (k_{mod} of 0.3 for permanent loads and k_{def} of 1.5 in service class 1), which, similar to stiffness, also shows a significant disadvantage compared to the other EWPs when it comes to long-term mechanical performance. Furthermore, Eurocode 5 [190] does not allow the use of OSB in service class 3. There are only few studies [192–198] on the long-term behaviour of OSB and often it is not

Table 3
Modification (k_{mod}) and deformation factor (k_{def}) of investigated EWPs as indicator for long-time performance under different climate conditions (service classes) based on Eurocode 5 [190]. Service class 1: Average moisture content does not exceed 12% for most softwoods (approximately 20 °C and 65% relative humidity for most of the year), Service class 2: Average moisture content does not exceed 20% for most softwoods (approximately 20 °C and 85% relative humidity for most of the year), Service class 3: Average moisture content higher than service class 2. The range represents load duration. *The range represents the respective load duration (permanent – instantaneous).

Limit state design principle	EWP	Service class 1	Service class 2	Service class 3	Ref.
k_{mod}^* , used in Ultimate limit state (ULS)	GLT	0.6–1.1	0.6–1.1	0.5–0.9	[190]
	CLT	0.6–1.1	0.6–1.1	0.5–0.9	[148]
	LVL	0.6–1.1	0.6–1.1	0.5–0.9	[190]
	PLY	0.6–1.1	0.6–1.1	0.5–0.9	[190]
	LSL	0.6–1.1	0.6–1.1	0.5–0.9	[191]
	OSB	0.3–1.1	0.3–0.9	–	[190]
k_{def} used in Serviceability limit state (SLS)	GLT	0.6	0.8	2.0	[190]
	CLT	0.6	0.8	2.0	[148]
	LVL	0.6	0.8	2.0	[190]
	PLY	0.8	1.0	2.5	[190]
	LSL	0.6	0.8	2.0	[191]
	OSB/3, OSB/4	1.5	2.25	–	[190]

specified which type of OSB (OSB/2, OSB/3 or OSB/4) is being studied. Therefore, the research and improvement of the long-term behaviour of EWPs in general and OSB in particular represents a current research gap which must be closed in order to make these materials fully competitive.

4.6. Introducing mechanical efficiency

The mechanical performances discussed in the previous chapters show that the established EWP underutilize the mechanical potential of (defect-free) wood to varying degrees. Additionally, the mechanical performance and the material yield of established engineered wood products were discussed but not related to each other so far. However, future decisions should consider a combination of resource efficiency as well as mechanical performance when comparing and selecting a material for a particular application. Therefore, a so-called “mechanical efficiency” is proposed in the next section. The mechanical efficiency is a synthesis of material yield and mechanical utilization and was calculated based on the following equation:

$$\text{Mechanical efficiency} = \frac{\text{Mechanical performance}}{\text{Mechanical potential}} \times \text{Process efficiency (i.e. material yield) [\%]} \quad (2)$$

where the mechanical performance is the approved mechanical property of the EWP (e.g. characteristic bending strength), the mechanical potential is the corresponding mechanical property of defect-free wood and the process efficiency is based on Eq. (1). The mechanical efficiency does not consider economic aspects as well as raw material availability, which are at least as important from an entrepreneurial point of view. However, both factors are heavily influenced by the current socioeconomic conditions and can shift within months if not weeks. Therefore, they are no pure indicator of technological performance and would rather introduce a volatile economic bias. For this reason, they were excluded from the mechanical efficiency in Eq. (2), but should be reintroduced in subsequent market studies.

With respect to the stiffness (Fig. 8A), an average (mean modulus of elasticity) of 12 GPa for softwoods (spruce, pine, fir and larch) and 12.2 GPa for hardwoods (oak, beech, birch, poplar and chestnut) was assumed for the mechanical potential based on established literature [132,180,181,199]. Considering the lumber-based EWP, the mechanical efficiency ranges from 27% for softwood CLT (35% for hardwood GLT) up to 44% for CLT as well as GLT based in hardwoods. The veneer-based materials achieve mechanical efficiency in the range of 38% for PLY (softwood and hardwood) up to 70% for hardwood LVL (59% for

softwood LVL). In regards to the strand-based EWPs, OSB achieves a mechanical efficiency of 40% while LSL is significantly higher at 66%. It must be added that the approved modulus of elasticity for EWPs is based on a mean value instead of a characteristic value, which generally increases the mechanical efficiency in contrast to the bending strength.

In regards to strength, the mechanical efficiency of the discussed EWPs was calculated assuming an average mean bending strength of 84 MPa for softwoods (spruce, pine, fir and larch) and 97 MPa for hardwoods (oak, beech, birch, poplar and chestnut) based on established literature [132,180,181,199]. The resulting mechanical efficiencies are depicted in Fig. 8B. The lumber-based EWPs show a mechanical efficiency from 9% for softwood CLT up to 16% for hardwood GLT, with hardwood CLT (14%) and softwood GLT (11%) in between. In comparison, the veneer-based products achieve a significantly higher mechanical efficiency ranging from 14% for softwood PLY (20% for hardwood PLY) up to 42% for hardwood LVL (25% for softwood LVL). The strand-based EWPs on the other hand are in between, achieving a mechanical efficiency of 20% for LSL and 23% for OSB.

Overall, the reported mechanical efficiency suggests a rather low utilization of the natural potential of wood and raises the question of whether the current processes are suitable for getting the optimum out of the raw material. In that regard it must also be mentioned that the current design and construction processes as well as methods in the industrialized countries focus on a high efficiency in the design and manufacturing to lower the personnel costs. This often leads to low material utilization which can be increased by structural optimization beside the discussed increases in the efficiency of the materials. In addition, the formability of the different EWPs in relation to their mechanical efficiency could be considered in a further step. Mechanical efficiency of a structural member can be achieved through the design of the cross-section high moment of inertia to cross-sectional area ratio, or shape factor [2]. Currently, most EWPs are manufactured in an undifferentiated shape with a constant rectangular cross-section. Depending on the fabrication techniques for the final structural products, the different EWPs will be more efficient at achieving higher shape factors. Furthermore, it has to be stated that currently widely used materials are characterized by a high inhomogeneity leading to a big scatter of the achievable material properties. This results in a comparatively high difference between the mean material properties and the required

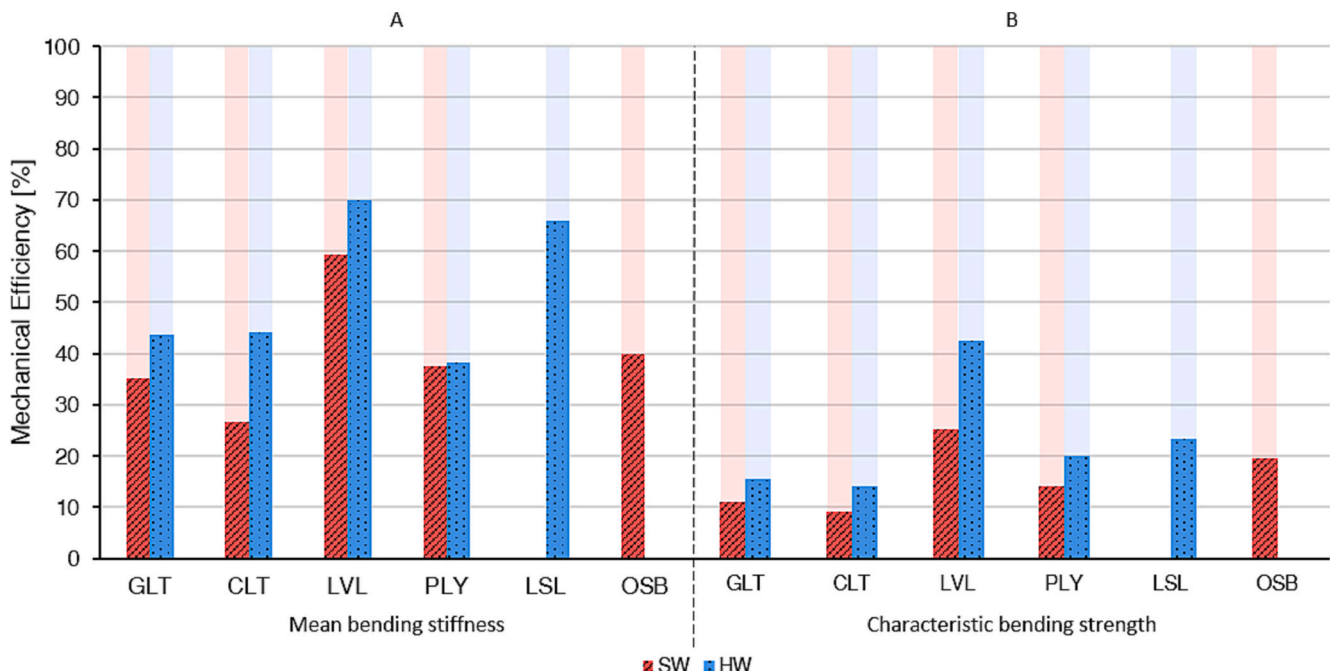


Fig. 8. Mechanical efficiency of established EWPs for construction based on mean bending stiffness (A) and characteristic bending strength (B).

properties to ensure the societally required safety level. Therefore, the mechanical efficiency could be increased by a better homogenization of the products and a finer resolution during characterisation leading to a smaller variation of the material properties. In addition, the variety in mechanical efficiency underlines the potential of hybrid engineered wood products for the building sector.

5. A concise look at environmental performance

Beside the mechanical performance of construction materials also the environmental performance plays an important role, especially in the face of climate change and the need to decarbonize the building sector. However, the evaluation of the environmental performance is a very complex topic including multiple possible approaches and indicators to consider [200]. As with long-term behaviour (see Section 4.5.), a detailed consideration of these aspects would go beyond the scope of this review. Therefore, selected LCA indicators [201,202] (functional unit is 1 m³ of the respective EWP) will be covered in this final section to provide a concise overview of the environmental benefit and impact of the investigated EWPs. The “Released carbon” and “Stored carbon” are used to indicate the net carbon storage potential (difference between “Stored carbon” and “Released carbon”) and the “Global warming potential” as well as the “Total primary energy” were selected to represent the environmental impact of 1 m³ EWP (see Table 4).

On average, 1 m³ of engineered wood product (cradle-to-gate) temporarily stores 953 kg CO₂ equiv. (260 kg C), releases 260 kg CO₂ equiv. (71 kg C) into the air, has a GWP of 216 kg CO₂ equiv. and needs 5569 MJ total primary energy during production. This would result in an average net carbon storage potential of 693 kg CO₂ equiv. (189 kg C), which is in line with assumptions made by Churkina et al. [12,17].

Taking a look at the EWPs based on their respective intermediaries, a few significant differences between the products can be seen. GLT (average material yield: 39%) and CLT (37%) have a net-carbon storage potential of 798 kg CO₂ equiv. (217 kg C), they have a GWP of 148 kg CO₂ equiv. and require 5074 MJ total primary energy for production. In contrast, LVL (52%) and PLY (52%) show a net-carbon storage potential of 657 kg CO₂ equiv. (179 kg C), while having a GWP of 232 kg CO₂ equiv. and requiring 7577 MJ total primary energy during production. LSL (69%) and OSB (83%) show a net-carbon storage potential of 664 kg CO₂ equiv. (181 kg C), they have a GWP of 270 kg CO₂ equiv. and OSB (no literature on LSL) needs 8114 MJ total primary energy during production. This tendency of an increasing environmental impact with decreasing size of the intermediary is well known in wood technology [33]. The disintegration of round wood into smaller intermediaries and the subsequent re-assembly to composites usually causes greater expenditure of energy and requires more auxiliary materials like synthetic adhesives. This further underlines the importance of resource

efficiency, as the amount of raw material needed to produce 1 m³ of EWP varies significantly and thus has a significant impact on the actual round wood demand but rather low impact on LCA indicators, as the provision of raw material is usually not that impactful. This is further illustrated in Fig. 9, comparing the average net carbon storage potential with the material yield.

Considering remarks on LCA indicators by Oneil et al. [55] it needs to be mentioned that the presented values in Table 4 and Fig. 9 need to be taken with a grain of salt for two main reasons. First, the values presented include two allocation principles (economic and mass allocation), with economic allocation usually leading to a higher environmental impact of the higher value product [55]. This significantly influences the outcome, as market prices of the main and co-products can fluctuate greatly over time. Second, the number of values found in literature varied greatly between the different products (see brackets in Table 4). This is especially critical when different production regions (e.g. varying transport distances) and actuality of the data (e.g. age of the mill) are considered.

Overall, the carbon storage potential and energy demand for production of the various EWPs have to be studied in more detail, also including additional regional differences. Furthermore, the underlying LCAs need to be kept up to date much more consistently, as factors such as environmental regulations, energy provision and efficiency, data quality and availability are subject to more frequent changes than, for example, mechanical performance and can significantly change the outcome of the assessment.

6. Summary and conclusions

Wood as a building material can contribute to reducing the share of energy-intensive construction to the necessary minimum. It thereby enables a secondary storage of CO₂ and while assuming sustainable forest management, will allow to contribute to a reduction of the total emitted CO₂. To accomplish this task, the finite resource wood must be used as efficiently as possible. Therefore, knowledge about the resource efficiency (i.e. the ratio between the output of intermediate or final product and raw material input) of different products apart from the structural efficiency is a key for future decisions. Ultimately decisions should be made based on the optimal performance in relation to the required raw material.

The present comparison of the resource efficiency of established structural, wood-based building materials (GLT, CLT, LVL, PLY, LSL and OSB) confirms a strong correlation between particle size (lumber, veneer or strand) and overall material yield. Based on the principal processes for first round wood breakdown (sawmilling, veneer peeling and stranding) and losses during subsequent processing the overall material yield increases with decreasing particle size. Based on 100% roundwood

Table 4

Selected indicators for the environmental performance of 1 m³ EWP. The numbers represent an average based on literature (see references in the table) dealing with LCA. In the case of stored carbon, a molecular weight ratio of 44/12 CO₂/C was assumed. The number in the bracket represents the number of values found in literature.

Indicator	Unit	GLT	CLT	LVL	PLY	LSL	OSB
Released carbon (cradle-to-gate)	[kg CO ₂ equiv.]	104 (5)	147 (3)	212 (5)	374 (2)	295 (1)	429 (1)
	[kg C]	28	40	58	102	80	117
Stored carbon (cradle-to-gate)	[kg CO ₂ equiv.]	931 (6)	886 (3)	952 (6)	866 (3)	1046 (1)	1033 (3)
	[kg C]	254	242	260	236	285	282
References	[-]	[58,113,114,203]	[116,118]	[123–125,204]	[126,204]	[129]	[56,126,205,206]
Global warming potential (GWP) (cradle-to-gate)	[kg CO ₂ equiv.]	156 (8)	139 (5)	303 (5)	161 (4)	295 (1)	244 (4)
Total primary energy (cradle-to-gate)	[MJ/m ³]	5560 (11)	4588 (2)	9275 (7)	5878 (6)	–	8114 (5)
Total primary energy (gate-to-gate)	[MJ/m ³] [-]	1310 (7)	580 (6)	4250 (7)	3133 (4)	13,386 (1)	4473 (6)
References		[55,58,113,114,203,207]	[57,116,118,119]	[55,124,125,204]	[55,126,208]	[129]	[56,126,205,206,209]

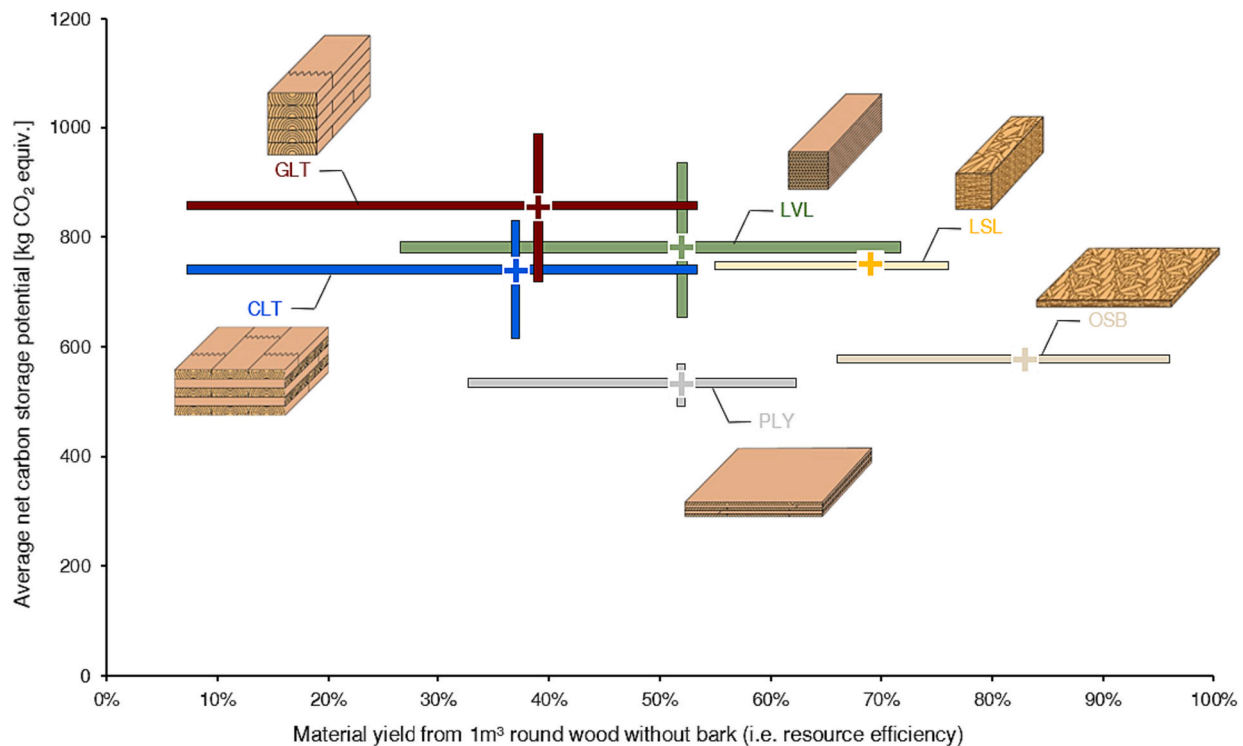


Fig. 9. Average net carbon storage potential [kg CO₂ equiv.] of 1 m³ of EWP based on values in Table 4 in relation to the achievable material yield [%] from 1m³ of round wood without bark as reported in Fig. 4 (total).

input without bark, the strand-based products allow roughly for a 69% (LSL) and 83% (OSB) output as primary target material, followed by the veneer-based products 52% (LVL) and 52% (PLY). Lastly, lumber-based products achieve an average output of 39% (GLT) and 37% (CLT). The variability in material yield for the production process of a product category is mainly driven by natural (mainly raw and intermediary material characteristics) and man-made factors (mainly market and process characteristics). In addition, the method used to determine either the volume or weight of round wood input or product output significantly influences the reported material yield. As the data was extracted from different sources this causes a certain residual inaccuracy when comparing different countries or products and must be considered especially when research is done using yield figures.

Based on currently available data, the influence of wood species on the process efficiency of sawmilling and veneer peeling seems to be negligible neglecting the raw material yield from a whole tree to the logs. This underlines the greater influence of the principal technology compared to the processed wood species. However, this requires a steady supply of similar logs as these mills are generally optimized towards certain raw material assortments. In regards to final material yield, a comparable statement about the influence of wood species cannot be made. Based on the reviewed literature it can be concluded that processing hardwoods leads to higher variability in achievable material yield. Since the literature on this topic was either limited or not available for certain products this needs to be addressed in more detail in upcoming research as the wood species composition will become more divers in the future.

The reviewed characteristic bending strength and modulus of elasticity demonstrated a wide range of mechanical performances over all products with varying degrees of corresponding resource efficiency. This proves that the structural potential of current engineered wood is given but an application according to the provided properties is essential. Therefore, the introduced mechanical efficiency aims to provide a benchmark for this material selection. Based on the reviewed literature it can be concluded that LVL (characteristic bending strength: 32% and

mean bending stiffness: 54%) and LSL (25% and 67%) offer the highest mechanical performance per input material, followed by OSB (19% and 40%), PLY (19% and 33%), GLT (14% and 40%) and CLT (12% and 27%).

From these benchmarks further improvements in mechanical efficiency can be discussed. For example, the use of hybrid components where materials with better mechanical performance are used in the higher stressed areas and vice versa could lead to a more sustainable utilization of wood. This is especially critical as the current trend is leaning more towards the least resource-efficient lumber-based products GLT and CLT [210]. However, especially structural and other technological (e.g. long-term behaviour, pre-fabrication and fasteners), physical (e.g. fire protection, acoustics, thermal insulation, airtightness and room climate), process (e.g. productivity, necessary pre-treatment and raw material flexibility), environmental (e.g. energy demand, transport distances and recyclability) as well as market (customer demand and perception) factors need to be considered to enable a sound decision on the best-suited product for each situation.

In summary, the shift towards climate-resilient cities requires a rethinking of the current structural design and material selection process with an added focus on resource efficiency. In this respect, the reviewed state of the art argues in favour of an increased use of veneer and strand-based wood products, which offer a trend towards an improved mechanical efficiency, in addition to currently mainly used lumber-based products. However, the long-term mechanical performance of these products and the environmental performance need to be considered as well. Overall it becomes clear that it will require a resource efficient combination of processes in order to satisfy the increasing demand for wood in construction.

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Declaration of Competing Interest

The authors declare that they have no conflicts of interest regarding this article.

Data availability

Data will be made available on request.

Appendix A. Appendix

Table A

Literature (Ref.) on the process efficiency of first round wood breakdown and yield of structural EWPs based on one unit [volume or weight] of raw material (round wood under bark, lumber or veneer) input. L.g = Lumber green, L.d&p = Lumber dried & planed, V.g = Veneer green, V.d&c = Veneer dried & clipped. LRR = Lumber recovery ratio, LRF = Lumber recovery factor, VRR = Veneer recovery ratio, VRF = Veneer recovery factor.

L.g	L.d&p	V.g	V.d&c	GLT	CLT	LVL	PLY	LSL	OSB	Basis	Ref.	Year
56%										LRR	[211]	1972
54%										$\frac{\text{kg lumber}}{\text{kg log}}$	[80]	1972
63%			52%									
45%	45%	77%	32%				45%			LRR, LRF, VRR, VRF	[70]	1976
63%			58%									
52%										$\frac{\text{ft}^3 \text{ lumber}}{\text{Int.0, 25 – inch logscale}}$	[212]	1978
63%										LRR	[213]	1980
52%												
37%										LRR, LRF	[63]	1984
45%												
38%												
71%												
74%										LRR	[87]	1990
59%										LRF	[78]	1991
		58%					48%					
		70%					55%					
		75%					57%			VRR	[214]	1991
		78%					59%					
46%												
47%												
52%												
55%												
42%										LRF	[60]	1992
58%												
42%												
53%												
	40%					52%		76%		% from log	[128]	1997
	42%						43%					
	51%						47%			% from log	[131]	2001
	60%						50%					
	64%	50%								LRR	[82]	2001
56%										LRF	[65]	2002
45%										LRR	[73]	2002
53%												
58%										LRR	[215]	2003
25%												
35%												
50%										LRR	[79]	2003
30%												
43%												
55%												
									81%	$\frac{\text{kg OSB}}{\text{kg log}}$	[206]	2005
				82%						$\frac{\text{m}^3 \text{ L.g}}{\text{m}^3 \text{ GLT}}$	[114]	2005
				82%						$\frac{\text{kg LVL}}{\text{kg veneer}}$	[125]	2005
						96%				LRR	[84]	2005
						91%				$\frac{\text{kg lumber}}{\text{m}^3 \text{ log}}$	[103]	2005
73%												
65%	57%											
62%	55%											
30%												
34%												
40%												
45%										LRR	[85]	2005
52%												
55%												

(continued on next page)

Table A (continued)

L.g	L.d&p	V.g	V.d&c	GLT	CLT	LVL	PLY	LSL	OSB	Basis	Ref.	Year
52%												
55%												
	41%											
	47%											
	51%											
	53%											
	56%											
	53%											
	52%						33%					
	44%						47%					
	51%						53%					
	53%						57%					
	55%						58%					
	58%						59%					
	55%						60%			LRR, LRF, VRR, VRF	[52]	2005
	53%						36%					
	39%						51%					
	44%						56%					
	49%						59%					
	50%						60%					
	51%						62%					
	50%						62%					
	41%											
	47%											
	51%											
	52%											
	53%											
	52%											
								73%	80%	% from log	[27]	2006
58%										LRR, LRF	[89]	2006
62%											[98]	2007
64%	44%											
									84%			
									76%			
									83%			
									83%	kg lumber	[71]	2007
									84%	kglog		
									80%			
									96%			
									66%	% from log	[216]	2008
		40%								VRR	[217]	2008
	73%											
45%						27%						
48%						36%						
49%						42%				LRR, % from log	[74]	2008
49%						46%						
50%						48%						
						54%						
39%										LRR	[86]	2008
39%									92%	kg OSB	[110]	2009
										kg log		
50%	39%									kg lumber	[99]	2010
										kglog		
41%												
42%										LRR, LRF	[75]	2010
44%												
46%												
52%												
55%										LRR	[88]	2010
57%												
59%	41%									m3 lumber	[100]	2012
		50%								kglog		
		43%								% from log	[106]	2013
		54%										
		61%								VRR	[218]	2013
63%												
57%										kg lumber	[90]	2013
										kglog		
				82%						m3 GLT	[203]	2013
										kg lumber		
					88%					m3 CLT	[116]	2013
										m3L.d&p		
59%	32%											
60%	29%									LRR	[77]	2013
49%	31%											

(continued on next page)

Table A (continued)

L.g	L.d&p	V.g	V.d&c	GLT	CLT	LVL	PLY	LSL	OSB	Basis	Ref.	Year
									75%	$\frac{\text{kg OSB}}{\text{kg log}}$	[205]	2014
		73%	54%							% from log	[107]	2014
								55%		$\frac{\text{kg LSL}}{\text{kg log}}$	[129]	2015
54%										LRR	[61]	2015
48%										LRR	[83]	2015
49%												
									75%	$\frac{\text{kg OSB}}{\text{kg log}}$	[209]	2015
	44%				43%					% from log	[62]	2015
	36%				53%					LRR	[81]	2016
48%												
40%										$\text{LRR, } \frac{\text{m}^3 \text{ L.d\&p}}{\text{m}^3 \text{ CLT}}$	[104]	2016
49%	32%				83%					$\frac{\text{m}^3 \text{ LVL}}{\text{kg V.d\&c}}$	[123]	2017
						88%				% from log	[126]	2017
							51%					
							58%		79%	$\frac{\text{m}^3 \text{ dried lumber}}{\text{m}^3 \text{ CLT}}$	[118]	2017
					83%					% from log	[219]	2017
									85%			
									95%	LRR	[59]	2017
65%	55%									LRR	[67]	2017
	50%											
	52%											
	47%											
	54%											
						89%				$\frac{\text{m}^3 \text{ LVL}}{\text{kg V.d\&c}}$	[123]	2017
56%				86%						$\frac{\text{kg lumber}}{\text{kg log}}$	[58]	2017
55%				88%						$\frac{\text{m}^3 \text{ dried lumber}}{\text{m}^3 \text{ GLT}}$	[53]	2017
59%										LRF	[220]	2017
									89%	% from log	[221]	2017
47%												
63%										LRF	[221]	2017
53%												
60%	47%									$\frac{\text{kg lumber}}{\text{kg log}}$	[102]	2018
					83%					$\frac{\text{m}^3 \text{ dried lumber}}{\text{m}^3 \text{ GLT}}$	[113]	2018
62%	49%									$\frac{\text{kg lumber}}{\text{kg log}}$	[101]	2018
										% loss from lumber	[115]	2019
				85%	85%					$\frac{\text{m}^3 \text{ lumber}}{\text{m}^3 \text{ CLT}}$	[119]	2019
					83%					LRR	[91]	2019
52%	33%											
51%	33%											
			55%						88%	% from log	[68]	2019
							55%		96%			
36%	9%				84%					% from log,	[57]	2020
50%	23%				83%					% losses from lumber		
64%	37%				82%							
					80%					$\frac{\text{m}^3 \text{ lumber}}{\text{m}^3 \text{ CLT}}$	[117]	2020
45%		75%								% from log	[222]	2020
64%	48%											
64%	44%						50%			% from log	[54]	2020
							44%					
		82%	75%							% from log	[108]	2021
		75%	69%									
		80%	71%									
	40%					53%	50%	70%	75%	% from log	[223]	2021

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